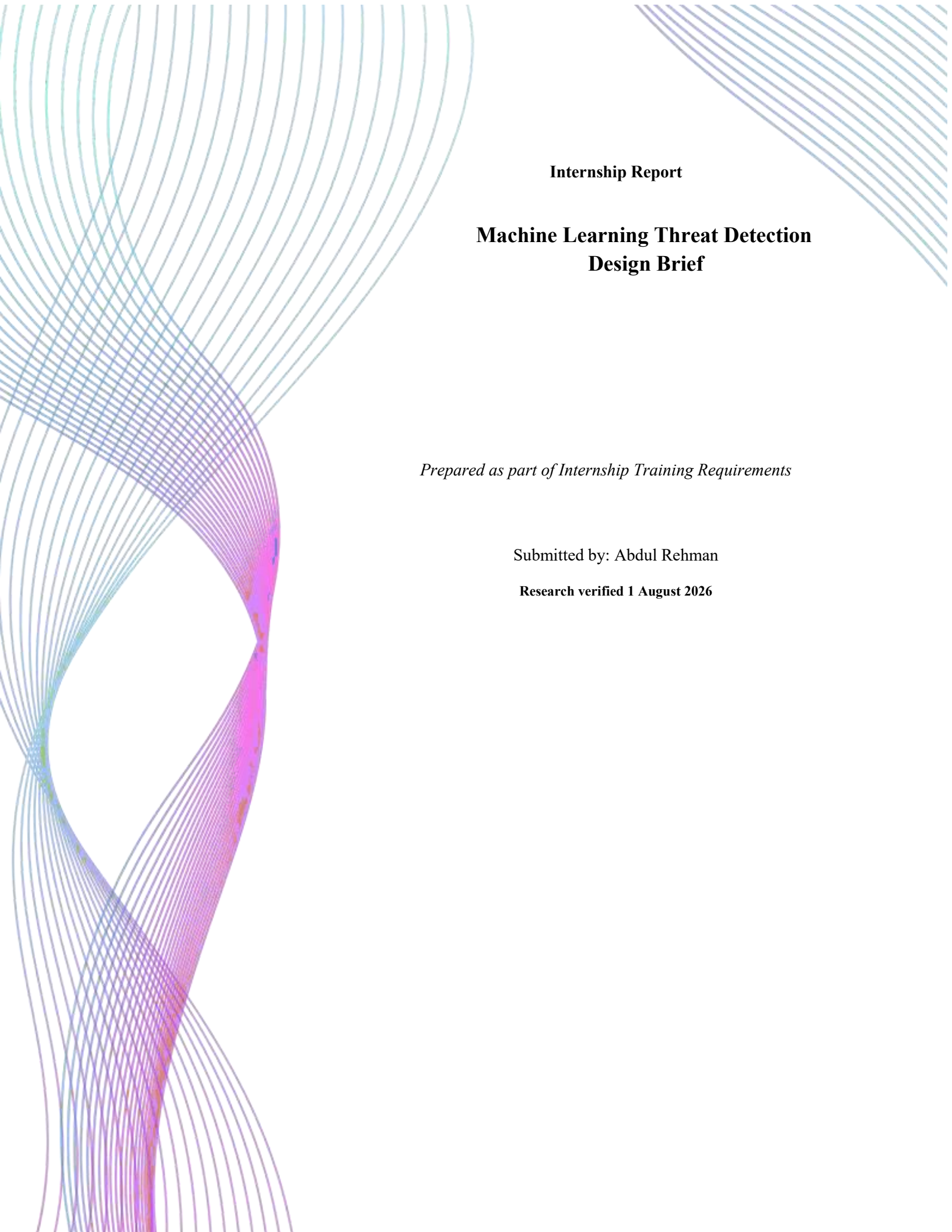




Internship Report

**Machine Learning Threat Detection
Design Brief**

Prepared as part of Internship Training Requirements

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Scenario: Network Intrusion Detection using the UNSW-NB15 Dataset

1. Problem Definition

Network intrusion detection aims to identify malicious network activity before it compromises systems or data. Modern attacks such as DoS, Exploits, Fuzzers, Reconnaissance, Backdoors, Shellcode, Worms and Generic attacks evolve rapidly, making signature-based intrusion detection systems insufficient for detecting previously unseen attack variants. Supervised machine learning learns behavioural patterns from historical labelled network traffic and can generalize to detect new malicious behaviour.

The objective of this project is to develop a machine learning model that classifies network traffic as benign or malicious with high reliability. In cybersecurity, false negatives are particularly dangerous because successful attacks remain undetected. False positives increase analyst workload and alert fatigue, so an effective model should maximize recall while maintaining strong precision.

2. Dataset Analysis

This project uses the UNSW-NB15 dataset (Kaggle version), derived from the UNSW Canberra Cyber Range. It is a widely used benchmark dataset for intrusion detection research and contains realistic modern network traffic with both normal and malicious activities. The commonly used version contains predefined training and testing sets suitable for supervised learning.

The dataset includes 49 engineered network-flow features describing packet behaviour, protocols, services, connection state, bytes transferred, durations and other statistical characteristics. Labels support both binary classification (Normal vs Attack) and multiclass classification through the `attack_cat` field, which contains nine attack categories including Analysis, Backdoor, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode and Worms.

Data preprocessing includes handling missing values, encoding categorical variables, scaling numerical features, checking duplicates and addressing class imbalance using techniques such as class weighting or SMOTE.

3. Model Selection

Algorithm	Why Selected	Strengths	Limitations
Random Forest	Baseline ensemble model	Robust, interpretable, feature importance	Can be slower on large forests
XGBoost	High-performance boosting	Excellent accuracy and handles imbalance well	Requires tuning
Neural Network	Captures complex nonlinear patterns	High predictive capacity	Less interpretable and computationally heavier

4. Evaluation Strategy

The dataset will be divided into training (70%), validation (15%) and testing (15%) sets using stratified sampling. Five-fold Stratified Cross-Validation will be used for hyperparameter tuning.

Performance will be measured using Accuracy, Precision, Recall, F1-Score, ROC-AUC and PR-AUC. Precision reflects the quality of generated alerts, while Recall measures how many attacks are successfully detected. Because missed attacks pose greater security risks than additional alerts, Recall and F1-Score are prioritised. ROC-AUC and PR-AUC provide threshold-independent evaluation, with PR-AUC being especially informative for imbalanced datasets.

5. Machine Learning Pipeline

Machine Learning Pipeline for Network Intrusion Detection (UNSW-NB15)



References (APA 7th)

- Moustafa, N., & Slay, J. (2015). UNSW-NB15: A comprehensive data set for network intrusion detection systems.
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<https://www.kaggle.com/datasets/mrwellsdavid/unsw-nb15>
- Scikit-learn Developers. User Guide: Classification Metrics and Model Evaluation.
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AI Assistance Note

ChatGPT was used to assist with literature organization and formatting.